Predicting Catastrophic Stock Implosions: A Machine Learning Approach

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Abstract

This paper introduces a new category of stock market crashes termed "Catastrophic Stock Implosions", characterised by significant price downturns followed by prolonged stagnation and minimal growth, making recovery highly improbable. By examining a universe of over 10,000 stocks and utilising more than 400 corporate fundamental features, we develop a machine learning model designed to predict such implosions before they occur. The final model demonstrates the ability to accurately identify over 60% of Catastrophic Implosions in the test dataset while maintaining a false positive rate of less than 3%. This robust detection framework offers a valuable tool for portfolio managers to mitigate the risks associated with these severe and unpredictable market events.

1 Introduction

Stock crashes - familiar yet fatal market disasters that are the bane of every investor's existence. Since 1968, the prediction of stock crashes and related firm-specific calamities, including bankruptcy and financial distress, has been a deep field of research. Yet from the great disparity in terms used across the space, it is evident that creating a distinct, practical metric for quantifying stock crashes remains challenging. Many studies focus less on the definition of a stock crash and more on the techniques used to predict it, leading to impractical results. This investigation addresses both the "what" and the "how" problems by introducing a specialised new category of stock price downturn called a *Catastrophic Stock Implosion* and leveraging a range of machine learning techniques to predict these movements before they occur, shielding investors against the most severe, yet often hidden, risks.

It is imperative to highlight that despite the broad range of groups affected by stock crashes (corporations, regulators, traders, etc.), the investigation is explored specifically from a portfolio manager perspective, with experiments carried out in collaboration with trading fund Banking Science. The motivation behind a Catastrophic Implosion is that it should encompass a downturn so severe that the likelihood of the stock recovering after is minimal - these stocks should be avoided entirely in a portfolio. While there is existing research into predicting financial distress, crash risk and bankruptcy, from an investor perspective none of these metrics prove truly ideal for modelling this type of movement. Prediction of bankruptcy, while valuable, does not always coincide with stock crashes which, for investors, is the real danger. As proven in the investigation, many stocks enter a *zombie* state where after crashing they linger in the market without being declared bankrupt, meaning investors are stuck with an illiquid asset until bankruptcy eventually follows or they are removed from the exchange. As highlighted in Figure 1 with Acorda, this can happen years later. On the other end of the spectrum, predicting stock crash risk is too sensitive, with many stocks often suffering from higher risk during periods of distress but quickly recovering afterwards. This is valuable for trading strategies that depend on the stock's recovery following a crash, but for detecting those stocks that are unable to recover, this does not prove to be as effective. Hence eliminating assets solely based on crash risk would dramatically narrow down one's portfolio. As a result, the study proposes a new type of crash called a Catastrophic Implosion as a middle ground between these two extremes. The definition constructed captures a distinct and dramatic decline in a stock's price that remains down for an extended duration. Figure 1 illustrates a comparison between the discussed measurements - note the frequency of stock crash risk and the lateness of bankruptcy date in comparison to the Catastrophic Implosion (highlighted). By labelling stocks according to whether they implode during their time series, a forecasting model was developed through supervised learning, enhancing risk management strategies for investment managers. In particular, the final model serves as a pre-screener, identifying equities at risk of 'catastrophically imploding' that can be eliminated or hedged against.

Naturally, the choice of predictor variables (features) to train the model with is crucial. Previous research has demonstrated strong correlations between weaker corporate fundamentals and

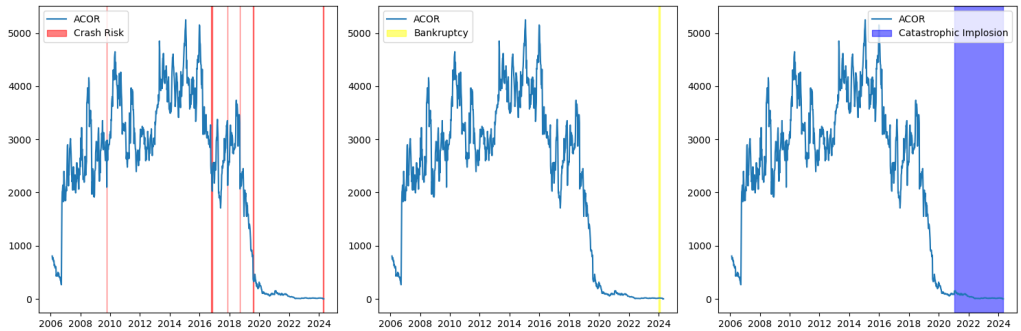


Figure 1: Stock crash risk vs Bankruptcy vs Catastrophic Implosion (Acorda Therapeutics)

higher price volatility/crash risk (Meng et al., 2023). The investigation reported here applies a similar approach, leveraging an array of over 400 fundamental and macroeconomic variables provided by Banking Science. A range of machine learning and deep learning models are trialled alongside extensive feature engineering and specialised hyperparameter tuning to detect as many implosions as possible. The results achieved indicate a positive response - the XGBoost model signalled over 60% of Catastrophic Implosions in the test set with a false positive rate of less than 3%, highlighting effectiveness in capturing the majority of implosions while avoiding a wipe-out of healthy stocks. Through feature importance analysis, the model relies heavily on deviations in earning yields, log returns and cash flow from operations. Such factors can prove to be valuable additions to further models and serve as suitable checkpoints for even deeper research.

The rest of the paper is organised as follows: Section 2 dives deeper into related fields including bankruptcy, financial distress and stock crash risk. Section 3 discusses the metrics designed for detecting Catastrophic Implosions from historical data and introduces a quantitative definition. Section 4 highlights the framework constructed for model development and training. Section 5 features a critical evaluation of the results, a description of refinement techniques performed and an analysis of feature importance post-testing. Lastly, Section 6 concludes the study and opens multiple doors for further research.

2 Background and Literature Review

The existing literature surrounding bankruptcy, stock crashes and corporate failure is complex due to the disparity in the definitions used. Bankruptcy prediction has remained a popular field of research since 1968, commencing with the Altman Z-score - a linear combination of valuation ratios that proved to be effective in signalling bankruptcy (Altman, 1968). Despite its occasional use today, Altman notes that the thresholds used at the time of release are now outdated and, with the large range in sizes of different companies, it can be difficult to apply universally. Nevertheless, Altman's work spurred research into the use of further statistical and eventually machine learning methods, including those demonstrated by Barboza et al. (2017), who found that models such as support vector machines and random forests enhanced bankruptcy prediction accuracy over the Z-score.

Yet despite beginning as a distinct field, predicting bankruptcy likelihood has become a subfield of more general 'financial distress' prediction, where a range of metrics are used. As noted by Shi and Li (2019), other terms employed include 'default firm', 'early-warning', 'failure' and 'insolvency'. Consequently, the array of dependent variables used is immense - some examples include distance to insolvency (Atkeson et al., 2017), negative trajectory in income/cash flow (Máté et al., 2023), loss in market value (McKibben, 2017) or even company claims that they were in 'dire straits about their payments to banks' (Altman et al., 1994). Surprisingly, despite such variation, the motivation behind these definitions does not tend to be a focal point in research. In many cases (Hacibedel and Qu, 2022) the labels for bankruptcy/distress are already present in the data set. Yet with almost every metric used, there are advantages and disadvantages - for instance, the date of bankruptcy might be concrete in definition but too late for a realistic prediction model. One may prefer to relax the criteria for a firm to qualify as 'distressed' (e.g., consecutive negative earnings/assets ratio), but this may lead to a less effective model. As a result, it is crucial to choose a variable that aligns with the goals of the model user, which, for this study, is the portfolio manager. As discussed in Section 3, the investor perspective is essential in motivating the definition of a Catastrophic Implosion. Hence our investigation contributes to the wider literature by introducing a unique metric for forecasting, highlighting its alignment for portfolio managers.

Our investigation also resembles the field of stock crash risk prediction. First introduced by

Hutton et al. (2009), stock crash risk is a distinct measurement of negative skewness in returns designed to indicate how likely a firm is going to suffer a drop in stock price at a given point in time. Contrary to bankruptcy and financial distress, the metrics for stock crash risk have remained largely consistent. The three main models include *CRASH*, *NCSKEW* and *DUVOL* (Habib et al., 2018). The metric examined in this study is *CRASH*, and is calculated by first regressing returns with the market index to calculate residuals and then applying a logarithm transformation. Weeks are labelled '1' or '0' according to the following:

$$CRASH = \begin{cases} 1 & w_{j,s} < \mu_y - 3.09\sigma_y \\ 0 & \text{otherwise.} \end{cases}$$

where σ_y and μ_y are the mean and standard deviation of the firm-specific returns $w_{j,s}$ for the year y respectively. As motivated previously, crash risk generally reflects distress in the price levels of a stock rather than complete collapses, often making it too frequent as a metric for Catastrophic Implosions. Indeed, as the metric focuses solely on the distribution of each stock's returns, most stocks suffer 'crashes' at least once during their lifespan, and generally recover shortly afterwards. This is in complete contrast to the proposed Catastrophic Implosion metric, which identifies only 6.6% of over 10,000 stocks as having imploded at least once. In addition to stock crash risk, there are many alternative approaches to modelling crashes, with one of the most popular being price bubbles (seeking an accumulation in price before a significant fall). However, these historically occur on a market/industry level as opposed to an individual stock basis (Schatz, 2020). Other metrics include looking out for when short-term moving averages cross long-term moving averages, as demonstrated by Naik and Mohan (2021).

Although Catastrophic Implosions are a distinct addition to the fields of bankruptcy, financial distress and stock crash risk, the experiments performed in our investigation are motivated by the latest research conducted in these related fields.

3 Implosion Definition and Metrics

The investigation proposes a two-part framework for modelling Catastrophic Implosions - the *initial crash* and the *zombie period*. The *initial crash* marks the point at which the implosion starts and would often suffice for typical stock crash metrics. At first, the following methods were proposed (inspired by existing literature):

1. Crash risk (*CRASH*) - a measurement of negative skewness in returns
2. Drawdown Identification - assessing significant downturns from peak-to-trough
3. Deviation from moving average - a check for a significant discrepancy from the (yearly) moving average

The problem with these methods is that they require an explicit drop in stock price in a short amount of time. While one could argue for this as a benefit, assuring a firm decrease, a Catastrophic Implosion should differ from a traditional stock crash in that a stock can implode *gradually* - rather than lose all market value in one week, a sufficient definition would also manage to capture significant losses over longer periods.

The proposed metric for the *initial crash* was chosen to be a drop in cumulative returns - for a given week should the cumulative gain dip below a specified threshold, then that *may* signify the start of a Catastrophic Implosion, depending on whether additional criteria are satisfied. Comparative analyses between the chosen methods and the prior draft methods highlight the approach's versatility, being able to detect implosions that would be identified by both the crash risk and drawdown techniques and those that were not detected by either. However, the metric is not without limitations. There may be a delay in reaching the threshold - for instance, a stock might experience a sharp decline, but it may not fall below the cumulative return threshold until much later, delaying the time that the implosion would be recorded to have started. In this study, this limitation is mitigated by making forecasts yearly, so the precise timing of the implosion is not critical.

The metric for measuring the *zombie period* was simpler to devise. After identifying the *initial crash*, the next step is to check whether the stock price eventually rises back above a certain threshold. If it does, the movement would not be classified as a Catastrophic Implosion, as the stock shows signs of recovery. However, suppose the price remains below the specified threshold for a sustained period (the *zombie period*). In that case, it indicates a more serious, long-term impact on the stock's value, suggesting the crash may have lasting effects on its future performance. Should the price remain at a sufficiently low level past a certain amount of time, the crash would

qualify as a Catastrophic Implosion. In Figures 1 and 2, the *zombie period* is represented by the shaded area.

Having defined the methodology for identifying a Catastrophic Implosion from historical data, it was necessary to set the specific thresholds for the following parameters:

1. C - the threshold of the cumulative returns that must be dropped below during the *initial crash*
2. M - the maximum cumulative return that cannot be surpassed during the *zombie period*
3. T - the length of the *zombie period* (in weeks)

This was done via a grid-based search strategy, running the algorithm across the stock universe with different combinations of parameters and assessing the implosions visually and by number.

Before discussing experimental results, it is necessary to describe the stock universe. The FactSet database (FactSet, 2023) provided by Banking Science consists of 10,826 US stocks between 2000-01-01 and 2023-11-02. A quarter of stocks belong to the Financial Conglomerates industry, with the rest being roughly uniformly distributed. For this experiment, weekly stock price data was monitored. The stock price data was adjusted for split and spin-off effects.

Exploring parameters was one of the more challenging aspects of the investigation, as there is no definitive 'correct' setting. After consulting with industry partners at Banking Science, it was determined that no more than 10% of the assessed stock universe should be classified as imploding. Based on this criterion, the following parameters were established:

1. $C = -0.8$
2. $M = -0.2$
3. $T = 78$

Weaker thresholds such as $C = -0.6$ and $M = 0.0$ were suggested first, however, we decided it was beneficial to be aggressive, ensuring that most imploded stocks followed the distinct, singular collapse movement we sought (shown in Figure 2). Furthermore, T needed not to be too large so that implosions across as many years as possible could be recorded (note in the methodology that the data is processed yearly). If, for example, $T = 104$, no implosions could be recorded in 2022, a year featuring record-high inflation in the US and several equity crises, as the time frame of the data set ends on 2023-11-02. Hence, parameters C and M were decreased instead. The setting also ensured that any stock that did implode only did so once, which proved to be very helpful during model development. Thus, the final definition for a Catastrophic Implosion is as follows:

A Catastrophic Stock Implosion occurs when the cumulative return of the stock falls below 80% and remains at least 20% below the price it fell to for a minimum of 78 weeks.

This definition captures the concept of a crash that results in the stock price being significantly lower than before for an extended period. To the author's knowledge, this is the first definition

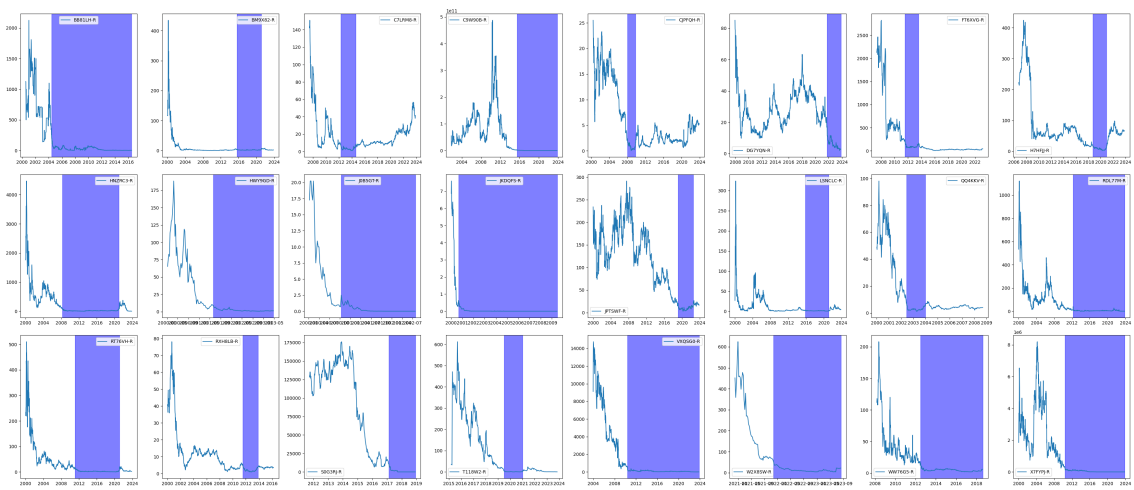
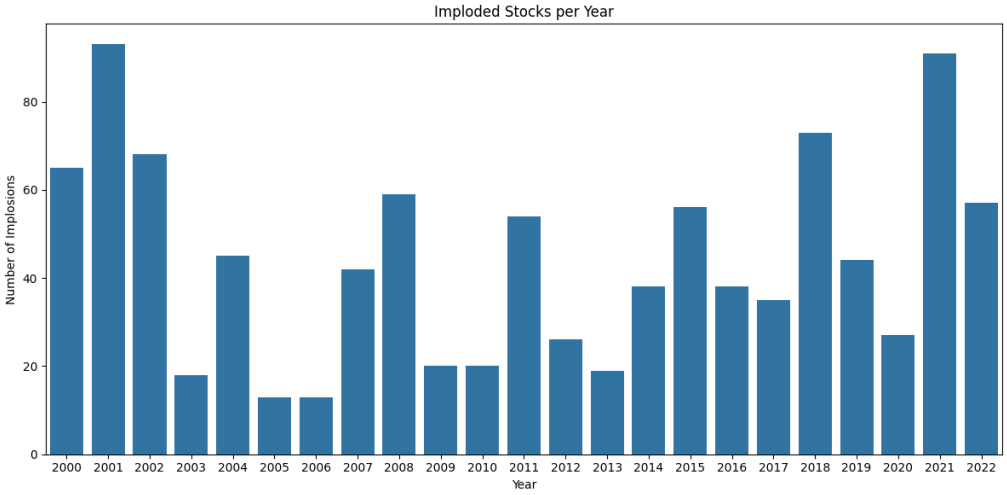
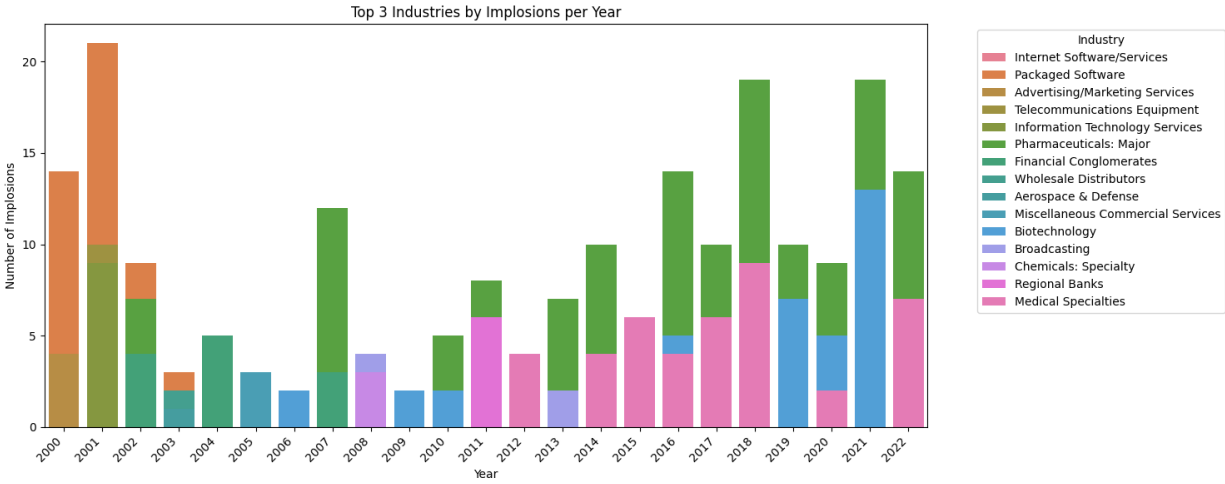


Figure 2: Sample of Implosions

that effectively encapsulates such a movement. As mentioned, the definition is not without limitations, such as how stocks might come close but not meet the exact specifications to qualify as



(a) Number of Implosions per Year



(b) Top 3 Industries with Most Implosions per Year

Figure 3: Number of Implosions

imploded - fortunately it is easy to adapt the parameters of the constructed algorithm to suit a specific risk appetite.

Figure 3 provides a breakdown of the number of implosions that occurred across the assessed timeline and the industries in which they occurred the most. Analysing implosions across years, 2001 and 2021 had the highest numbers of implosions. It is evident upon industry examination that the spike in implosions in 2001 was the result of the Dot-com Bubble. In 2021, the rise is attributed to the pharmaceutical and biotechnology sectors. Pharmaceutical companies consistently suffered throughout history, largely due to the inherent risks that come with drug development, including losing rights to medications shortly after development (Bell, 2024). However, the biotechnology spike was triggered by the COVID-19 pandemic, in which early-stage companies began with high valuations but rapidly lost money as demand for vaccines sharply fell (Megaw, 2023). Megaw noted that 'over 250 healthcare companies listed in the US between 2020 and 2021' despite them not being 'ready to be public'. These analyses provide initial insights into the potential factors behind Catastrophic Implosions.

4 Methodology

Having formed the definition of a Catastrophic Implosion, the next step was to build a forecasting model. A supervised learning approach was chosen, leveraging a binary dependent variable based on the proposed definition and an array of fundamental and macroeconomic variables. As highlighted previously, corporate fundamentals have proven to be valuable for price movement prediction in several studies, suggesting that firms displaying weaker balance sheets and accounting variables are more likely to suffer from crashes and insolvency. By assessing the performance of the final model, one can establish whether a similar relationship between fundamentals and Catastrophic Implosions can also be substantiated. Before commencing model development, however, several critical factors must be considered, one of the most important being the *which vs when* problem.

As the model analyses thousands of stocks, each with their own time series, it is attempting two problems - identifying *which* stocks across the portfolio are going to implode and *when* in their time series the implosion is going to occur. It is crucial to highlight that, from an investor perspective, the *which* sub-problem is much more important than the *when* sub-problem - it is more useful for the investor to correctly identify as many imploding stocks as possible without always getting the timing of the implosion correct than to identify a handful of implosions with accurate timing. Such a prioritisation plays an important part in both the construction and evaluation of the model.

To tackle the problem via a supervised learning framework, the following dependent variable is used:

$$y_{s,t} = \begin{cases} 1 & \text{if } s \text{ implodes in year } t+1 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where t represents a year in time and $y_{s,t}$ represents the target variable for stock s in year t. Each sample $\mathbf{x}_{s,t}$ consists of a vector representing independent, fundamental/macroeconomic variables x_i at the current time t for a given stock s. Hence, the model works across stocks at a yearly frequency, tackling both the *which* and *when* sub-problems while still prioritising the former, and ensuring effectiveness for the trader. Since investors care about avoiding Catastrophic Implosions entirely, the *zombie periods* following any implosions were not included in the data set, only the year that the implosion starts or the full available time series if the stock does not implode at all. For the feature set, the FactSet database provided by Banking Science offers over 1000 income, debt, cash flow, costs, margins, value and assets variables associated with each company. Macroeconomic variables (GDP, CPI, unemployment rate) were pulled from the FRED database (Federal Reserve Bank of St. Louis, 2024). Due to the overwhelming number of variables, several feature selection tools were employed, including the Boruta algorithm (Ali, 2022). The resulting feature set is shown in Table 1.

Feature	**Definition**
ff_assets_per_emp	Assets Per Employee
ff_cash_curr_assets	Cash & Equiv % Current Assets
ff_cash_div_cf	Cash Dividends/Cash Flow
ff_cf_ps_gr	Cash Flow Per Share - 1 Year Growth
ff_cf_sales	Cash Flow to Sales
ff_dfd_tax_assets_lt	Deferred Income Tax - Non-Current Asset
ff_div_yld	Dividend Yield
ff_earn_yld	Earnings Yield
ff_ebit_oper_roa	Return on Average Assets (EBIT)
ff_eff_int_rate	Interest Rate - Effective
ff_emp_gr	Employees - Total - 1 Yr Growth
ff_entrpr_val_sales	Enterprise Value to Sales
ff_for_sales_pct	Foreign Sales % Total Geographic Segment Sales
ff_intang_oth	Total Net Other Intangibles
ff_invest_st_tot	Total Short-Term Investments
ff_min_int_tcap	Accumulated Minority Interest % Total Capital
ff_mkt_val_gr	Market Value - 1 Yr Growth
ff_net_inc_per_emp	Net Income/Employee - Most Recent Fiscal Year
ff_oper_ps_net_cf	Cash Flow from Oper Per Share - Net
ff_reinvest_rate	Reinvestment Rate
ff_roic	Return on Average Invested Capital
ff_shs_float	Float Shares
ff_xord	Extraordinary Items
ff_std_debt	Short Term Debt % Total Debt
ff_compr_inc	Comprehensive Income
GDP	Gross domestic product
Unemployment_Rate	Unemployment Rate
log_return	Log Return

Table 1: Features Selected via Boruta (FactSet (2023), Federal Reserve Bank of St. Louis (2024))

Another crucial factor to be considered was how much historical data should be used for each prediction. Related literature on stock crash risk suggests the answer is non-trivial, thus two separate experiments were conducted, one incorporating a 1-year rolling window and the other a 5-year rolling window. For Approach 1 (1-year rolling window) minimal feature engineering was required as aggregated yearly data could be simply pulled from the provided database. Approach 2 (5-year

rolling window) required significantly more engineering, as each sample had to encompass 5 years' worth of fundamental data. Note that this also meant only stocks with at least 5 years' worth of fundamental history could be included, reducing the universe size from 10,000 to 7000 for this approach. A popular solution for encompassing rolling windows is to create additional lag features. However, for a data set with a yearly frequency, this would result in an immense number of null values (one lag alone would introduce 7000 null values, one for each stock). The proposed solution was to perform aggregation using an open-source time series transformation tool called TSFEL (Time Series Feature Extraction Library) (Barandas et al., 2020). The library efficiently calculates statistical, temporal and spectral aggregated features at a large scale. The tool was applied to

Statistical	**Temporal**
Absolute Energy	Area Under Curve
ECDF Percentile Count	Autocorrelation
Entropy	Mean Absolute Diff
Histogram	Median Absolute Diff
Max	Mean Diff
Mean	Median Diff
Median	Negative Turning Points
Median Absolute Deviation	Positive Turning Points
Min	Sum Absolute Diff
Root Mean Square	
Stdev	
Variance	

Table 2: Chosen TSFEL transformations

each stock over a 5-year rolling window using the transformations shown in Table 2, resulting in a final set of over 400 transformed features.

The final stage in the core methodology was the model training. For this investigation, both classical machine learning algorithms (logistic regression, random forest, gradient boosting) and deep learning algorithms (Multilayer Perceptron) were experimented with. Training was performed on a Hadoop Distributed File System cluster, distributed across 12 nodes where possible. Since this required the PySpark model implementations over the typical scikit-learn versions, some algorithms could not be distributed. In fact, the scikit-learn and PySpark implementations yielded very distinct results from one another. It was discovered that this was the result of restrictions on the hyperparameter space during optimisation for PySpark models. For the 5-year rolling window, the training period spanned from January 31, 2004, to December 31, 2018, with the validation set within the training set (from January 31, 2017, to December 31, 2018) and testing conducted on data from January 31, 2019, to November 2, 2023. This resulted in an 80/20 split between the training and testing data (the 1-year rolling window approach was split similarly).

5 Model Evaluation

Before diving into results and feature analysis, it is valuable to revisit the objectives of the model from an investor perspective. While it's crucial to strike a balance between precision and recall, prioritising recall is more critical when predicting Catastrophic Implosions. The rationale is straightforward: the consequences of mistakenly removing a healthy stock from a portfolio are far less severe than those of failing to identify a stock that does implode. Moreover, linking back to the *which vs when* problem, one should evaluate models more by how many imploding stocks are detected and less by how many are detected at the correct point in time. For instance, if the model predicts an implosion next year that ends up occurring in two years, the prediction is naturally still valuable - it is crucial to remember that the metrics utilised for detecting Catastrophic Implosions do not always align with the true starting point of the downturn.

The initial results from training and testing the vanilla models are not very impressive - the models have an easy time classifying healthy periods but, in some cases, can barely detect any implosions (not ideal for an implosion prediction model!). To fix this, the models must be adapted to tackle the class imbalance. This can be done through class weighting - applying larger weights to the loss function of the algorithm to penalise the misclassification of positive samples more heavily. In addition, for targeting a stronger recall over precision, hyperparameter tuning is essential. This investigation employed Bayesian Optimisation for hyperparameter tuning, using the MCC as the objective function to ensure the model strives for accurate results across all quarters of the confusion matrix. As well as the standard hyperparameters for each model, where possible class weighting was also tuned during the optimisation.

Results from Approach 2 (5-year rolling window) outperform those attained from Approach 1

Table 3: Model Evaluation Results (Approach 2 - 5-year rolling window)

Model	Precision (0)	Precision (1)	Recall (0)	Recall (1)	F1 Score (0)	F1 Score (1)	MCC
LR	1.00	0.06	0.88	0.80	0.94	0.11	0.21
RF	1.00	0.21	0.99	0.46	0.99	0.30	0.31
GBT	1.00	0.08	0.92	0.79	0.96	0.15	0.24
XGB	**1.00**	**0.20**	**0.99**	**0.61**	**0.99**	**0.33**	**0.35**
MLP	1.00	0.14	0.97	0.50	0.98	0.22	0.26

(1-year rolling window) across all metrics, suggesting the importance of incorporating longer-term historical data for prediction. For Approach 2, the best performer was the XGBoost model, attaining the highest MCC of 0.35 and skewing in the direction of recall, predicting 61% of implosions that occurred in the test set. Although the low precision suggests a high number of false positives, the AUC score of 0.95 and false positive rate of 2.41% highlight that the number of false positives relative to the full data set is very small. Of course, as the first investigation into Catastrophic Implosions, it is impossible to benchmark these results against other research, however, consultation with Banking Science confirmed the model's robustness in distinguishing implosions across the stock universe. Furthermore, if one chooses to focus entirely on the stocks that were detected to implode, disregarding the model's yearly predictions of when they would occur, the MCC score raises to 0.44, further supporting the theory that the model is capable of identifying stocks likely to implode but may need further refinement to improve its timing. One can even examine the false positives to recognise that although they may not fit the definition of a Catastrophic Implosion, they still suffer from great falls in share value.

Taking a closer look at the Gini importances and SHAP values derived from the XGBoost model,

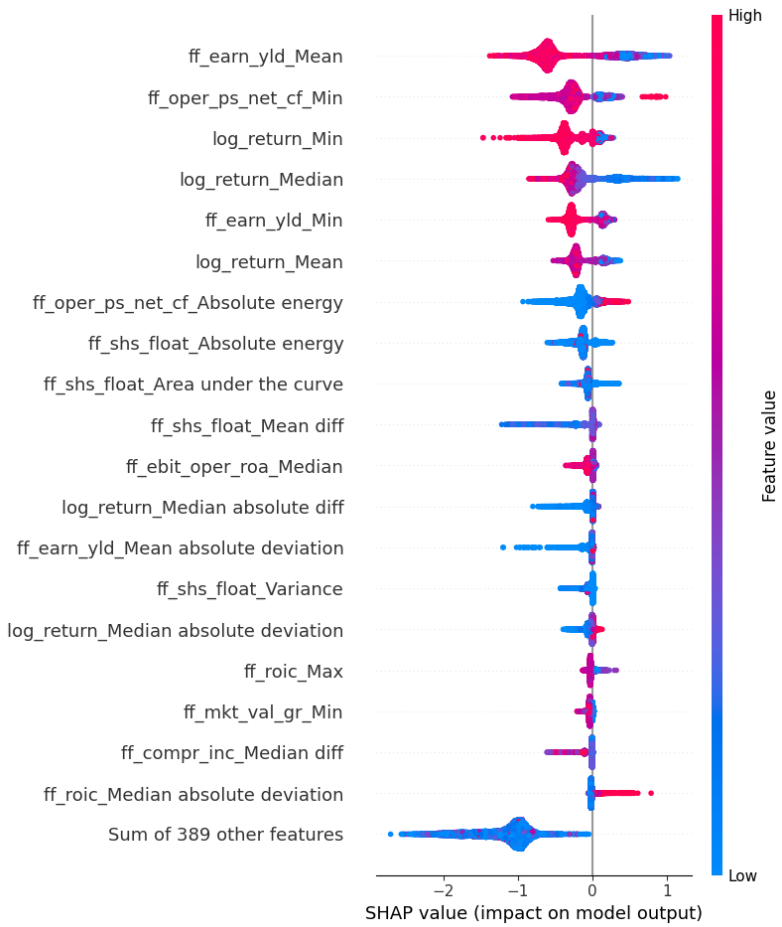


Figure 4: XGBoost SHAP Beeswarm (5-year Rolling Window)

earning yields, operations cash flow and log returns were found to be the top features driving model predictions. The SHAP values (Figure 4) provide an insight into the model's decision-making - for example, a higher average earnings yield over the previous 5 years pushes the model away from signalling an implosion. Such results align with existing related literature (Altman (1968), Meng et al.

(2023), Jiang et al. (2024)). Similarly, transformations including the mean and minimum prove to be the most frequent among the top features, also signifying their use in future investigations and models.

6 Conclusions

In constructing a quantitative definition for a Catastrophic Implosion that captures the notion of a collapse followed by long-term stagnation and then developing a machine learning model to forecast such events, the investigation achieved the main objectives originally set. A Catastrophic Implosion was defined to be a crash where the cumulative returns fall below -0.8 and do not rise by more than 20% for at least 78 weeks following, with visualisations proving its alignment to the type of movement being sought for in this study. Using this as the basis of a binary dependent variable, a range of machine learning models were then trained across 7000-10000 stocks (depending on the rolling window size), each with over 400 time series features, to predict implosions one year in advance. The experiment was first performed using a 1-year rolling window of previous data and then a 5-year window, with results confirming the latter to be more effective. XGBoost proved to be the most versatile option in implosion detection, predicting 61% of implosions in the test set with a false positive rate of less than 3%, demonstrating robust ability in signalling Catastrophic Implosions while also leaving room for improvement. Finally, feature importances suggest the value of cash flow from operations, earning yields and log returns in predicting Catastrophic Implosions, aligning with related studies (Altman (1968), Li et al. (2024)).

As the first to delve into Catastrophic Implosions, the study welcomes further research. Future work could extend the current definition to other asset classes, such as bonds, or explore entirely different approaches to modelling severe downturns. The challenge of defining criteria that capture both sudden and prolonged crashes remains an open problem. One might suggest using a multi-criteria definition, as proposed by Jiang and Jones (2018) for financial distress, but this could also complicate prediction efforts. In addition, models such as LSTMs and Transformers remain to be experimented with - this study focused primarily on adapting existing models to the problem at hand, rather than exhaustively testing every possible option. However, given LSTMs' state-of-the-art performance in time series forecasting (Han and Fu, 2023), incorporating them could potentially enhance the results. Finally, there is a promising avenue for research into unsupervised learning methods - the rarity of imploded stocks would suggest modelling them as 'anomalies' in a larger universe. A fair amount of engineering may be required, however, to ensure models such as isolation forests or autoencoders detect the 'correct' type of anomalies, as stocks can exhibit extremely volatile behaviour. Overall, the investigation opens several pathways for demystifying and forecasting Catastrophic Stock Implosions and thus a new avenue for enhancing risk mitigation strategies and bolstering dynamic portfolio optimisation.

References

Ali, M. (2022). Boruta feature selection explained in python. `https://medium.com/geekculture/boruta-feature-selection-explained-in-python-7ae8bf4aa1e7`. Medium.

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The journal of finance*, 23(4):589–609.

Altman, E. I., Marco, G., and Varetto, F. (1994). Corporate distress diagnosis: Comparisons using linear discriminant analysis and neural networks (the italian experience). *Journal of banking & finance*, 18(3):505–529.

Atkeson, A. G., Eisfeldt, A. L., and Weill, P.-O. (2017). Measuring the financial soundness of us firms, 1926–2012. *Research in Economics*, 71(3):613–635.

Barandas, M., Folgado, D., Fernandes, L., Santos, S., Abreu, M., Bota, P., Liu, H., Schultz, T., and Gamboa, H. (2020). Tsfel: Time series feature extraction library. *SoftwareX*, 11:100456.

Barboza, F., Kimura, H., and Altman, E. (2017). Machine learning models and bankruptcy prediction. *Expert Systems with Applications*, 83:405–417.

Bell, J. (2024). Acorda files for bankruptcy, reveals asset sale plan. `https://www.biopharmadive.com/news/acorda-bankruptcy-asset-sale-merz-stalking-horse/712010/`. Biopharma Dive.

FactSet (2023). Apple inc. financial data. FactSet.

Federal Reserve Bank of St. Louis (2024). Federal reserve economic data (fred).

Habib, A., Hasan, M. M., and Jiang, H. (2018). Stock price crash risk: review of the empirical literature. *Accounting & Finance*, 58:211–251.

Hacibedel, M. B. and Qu, R. (2022). *Understanding and predicting systemic corporate distress: a machine-learning approach.* International Monetary Fund.

Han, C. and Fu, X. (2023). Challenge and opportunity: deep learning-based stock price prediction by using bi-directional lstm model. *Frontiers in Business, Economics and Management*, 8(2):51–54.

Hutton, A. P., Marcus, A. J., and Tehranian, H. (2009). Opaque financial reports, r2, and crash risk. *Journal of financial Economics*, 94(1):67–86.

Jiang, F., Ma, T., and Zhu, F. (2024). Fundamental characteristics, machine learning, and stock price crash risk. *Journal of Financial Markets*, page 100908.

Jiang, Y. and Jones, S. (2018). Corporate distress prediction in china: A machine learning approach. *Accounting & Finance*, 58(4):1063–1109.

Li, Y., Xue, H., Wei, S., Wang, R., and Liu, F. (2024). A machine learning approach for investigating the determinants of stock price crash risk: Exploiting firm and ceo characteristics. *Systems*, 12(5):143.

Máté, D., Raza, H., and Ahmad, I. (2023). Comparative analysis of machine learning models for bankruptcy prediction in the context of pakistani companies. *Risks*, 11(10):176.

McKibben, M. (2017). *Predicting bankruptcy and catastrophic loss: A portfolio approach.* PhD thesis, Duquesne University.

Megaw, N. (2023). Biotech stocks undergo recovery after year of 'wreckage'.

Meng, Y., Shen, D., and Xiong, X. (2023). When stock price crash risk meets fundamentals. *Research in International Business and Finance*, 65:101975.

Naik, N. and Mohan, B. R. (2021). Novel stock crisis prediction technique—a study on indian stock market. *IEEE Access*, 9:86230–86242.

Schatz, M. (2020). *Financial Modeling of Bubbles and Crashes.* PhD thesis, ETH Zurich.

Shi, Y. and Li, X. (2019). An overview of bankruptcy prediction models for corporate firms: A systematic literature review. *Intangible Capital*, 15(2):114–127.